PH-1010 (January - May 2026)

Time:	(**Slot-D**) 11:00 – 11:50 (Monday); 10:00 – 10:50 (Tuesday); 09:00 – 09:50 (Wednesday); 13:00 – 13:50 (Thursday).
Place:	See Batch allocation .

Grading Policy: Quiz-I (20 Marks), Quiz-II (20 Marks), End Sem (60 Marks).

Important Dates:

Quiz-I ..... February 26, 2026
Quiz-II ..... March 19, 2026
End Sem ..... May 11, 2026

Class Policy: Regular attendance is essential and expected. Active participation during tutorial sessions are encouraged.
Course Pages: Tutorial problem sets, solutions, and additional updates will be made available on the Moodle page.

References: This is a list of various interesting and useful text and reference books that will be consulted during the course.

Textbooks:

1. J.R. Taylor, *Classical Mechanics*.
2. D. Morin, *Introduction to Classical Mechanics*.
3. T. Thornton & J.B. Marion, *Classical Dynamics of Particles and Systems*.
4. H.M. Schey, *Div, Grad, Curl, and All That*.

Additional references:

- D. Kleppner and R. Kolenkow, *An Introduction to Mechanics*,
- C. Kittel, W. D. Knight, M. A. Ruderman, C. A. Helmholz and B. J. Moyer, Mechanics, *Berkeley Physics Course, Volume-I*,
- R. P. Feynman, R. B. Leighton and M. Sands, *The Feynman Lectures on Physics, Volume I.*

Learning outcomes: Upon successful completion of this course, students will be able to

- apply Newton's laws to model translational and rotational motion, including motion in curvilinear coordinates;
- solve oscillation problems, including damped and driven systems, and analyze phase-space trajectories, with applications to nonlinear oscillations;
- understand the concept of potential functions and effectively apply conservation principles;
- understand the Lagrangian formulation and use it to analyze motion under central forces and interpret orbital dynamics;
- apply concepts from vector calculus to the study of fields and fluid mechanics.

Syllabus and Course Plan

Module 1: Introduction to Vectors (4 Lectures + 1 Tutorial)

(i)	Fundamental concepts: space, time, mass, and the point particle – Newton's axioms of motion, Vectors and coordinate systems – differentiation of vectors: velocity and acceleration	*Taylor 1.1 - 1.3*
(ii)	Transformation properties of vectors - the rotation matrix	*Thornton-Marion 1.2 - 1.3*
(iii)	Vector products: Dot product, Summation convention, orthogonality (Kronecker delta δ_{ij}); Cross-product of two vectors, the Levi-Cevita symbol (ϵ_{ijk}).	*Thornton,Marion 1.10,1.11, 1.12*

Module 2: Newton's Laws of Motion (5 Lectures + 1 Tutorial)

(i)	Newton's first and second laws as differential equations; inertial and non-inertial frames	*Taylor 1.4*
(ii)	Pseudo-forces in accelerating frames; uniformly accelerating systems	*Taylor 9.1*
(iii)	Newton's third law, conservation of momentum, validity of third law and example(s) of breakdown	*Taylor 1.5*
(iv)	Newton's second law in two-dimensional polar co-ordinates with examples - sliding block, planar and circular pendulum, oscillating skateboard	*Taylor 1.6 - 1.7*

Module 3: Oscillations(5 Lectures + 1 Tutorial)

(i)	Simple harmonic motion — equation of motion and explicit solutions. Energy considerations in oscillatory systems and small oscillation	*Taylor 5.1*
(ii)	Damped oscillations: underdamped, overdamped, critically damped cases	*Taylor 5.4*
(iii)	Driven harmonic motion and resonance phenomena, LCR circuits	*Taylor 5.5 - 5.6*

14 Lectures + 3 Tutorials + 1 Backups/Revision = 18 Teaching Days

Quiz-I

Module 4: Motion in 1D, Phase Space (8 Lectures + 1 Tutorial)

(i)	Kinetic and potential energy; work-energy theorem in one dimensions	*Taylor 4.1, 4.2*
(ii)	Energy in 1-d systems: Time period, Energy equation, Periodic motion, Turning points, Stable and unstable equilibrium points.	*Taylor 4.6*
(iii)	Introduction to phase-space. Phase space of harmonic oscillators- undamped and damped.	*Thornton and Marion 3.4, 3.5*
(iv)	Phase diagram for nonlinear systems, plane pendulum.	*Thornton and Marion 4.3, 4.4, Appendix B.2*

Module 5: Conservative Forces (4 Lectures + 1 Tutorial)

(i)	Work-energy theorem in higher dimensions.	*Taylor 4.3*
(ii)	Force as the gradient of potential energy — conservative and non-conservative forces in Cartesian co-ordinates only.	*Taylor 4.4*
(iii)	Geometric interpretation of gradient; properties of line integrals. Curl of a vector field and the condition for conservativeness (Cartesian only).	*Taylor 4.4*
(iv)	* Line element, gradient and kinetic energy in Curvilinear co-ordinates (cylindrical and spherical).	*Taylor 4.8*

12 Lectures + 2 Tutorials + 1 Quiz discussion + 1 Backups/Revision = 16 Teaching Days

Quiz-II

∗This part will be considered in End-Sem only.

Module 6: The Lagrangian Formalism (4 Lectures + 1 Tutorial)

(i)	Generalize coordinates, Euler-Lagrange equations,	*Morin 6.1, 6.3*
(ii)	Cyclic co-ordinates, conservation laws.	*Morin 6.4, 6.5*

Module 7: Motion Under Central Forces (6 Lectures + 1 Tutorial)

(i)	Angular momentum and its conservation. Lagrangian in plane polar coordinates — effective potential	*Morin 7.1, 7.2*
(ii)	Equation of orbit, bounded and unbounded orbits, planetary motion and Kepler's laws.	*Taylor 8.5, 8.6*
(iii)	Centre of mass, Reduced mass and two-body problem	*Taylor 8.1, 8.2*
(iv)	Satellites, transfer of orbits	*Taylor 8.8*

Module 8: Vector Calculus (6 Lectures + 1 Tutorial)

(i)	Flux of a vector field, Gauss's divergence theorem. Divergence theorem in cylindrical and spherical co-ordinates.	*Schey Chapter II*
(ii)	Stokes's theorem and circulation and curl of vector fields	*Schey Chapter III*
(iii)	Fluid flow, continuity equation, and Bernoulli's principle	Taylor 16.12

16 Lectures + 3 Tutorials + 1 Quiz discussion + 2 Revision = 22 Teaching Days

End-Sem